

used for silicon chips. That makes it small enough to fit into smartphones.

Higher frequencies mean shorter wavelengths and hence smaller antennas. The antenna actually becomes cheaper with the smaller size because it needs less silicon. There is a gas plasma alternative but it's not solid-state, so it is bigger and contains moving parts—making it more of a pain to manufacture. That leaves the door open for solid-state plasma antenna to be used for next generation Wi-Gig (its version 1.0 was announced in December 2009) that can reach up to 7Gbps bandwidth over frequencies up to 60 GHz.

Development progress

Initial investigations were related to the feasibility of plasma antennas as low-radar cross-section radiating elements with further development and future commercialisation of this technology. The plasma antenna R&D project has proceeded to develop a new antenna solution that minimises antenna-detectability by radar at the first instance. But since then an investigation of the wider technical issues of existing antenna systems has revealed areas where plasma antennas might be useful.

A significant progress has been made in developing plasma antennas. Present plasma antennas have been operating in the region of 1 to 10 GHz. Field trials have shown that an energised plasma reflector is essentially as effective as a metal reflector. However, when de-energised, the reflected signal drops by over 20 dB. Still some technicalities related to plasma antennas like increasing the operating plasma density without overloading the plasma discharge tubes, reducing the power required and the plasma noise caused by the ionising power supply, etc, have to be looked into in order to make them the useful technologies for wireless communication in near future.

The future of high-frequency, high-speed wireless communications

could very well be plasma antennas capable of transmitting focused radio waves that would quickly dissipate using conventional antennas. Thus, plasma antennas might be able to revolutionise not just high-speed wireless communications but also radar arrays and directed energy weapons. The good news is that plasma antennas will be on-the-shelf in the next couple of years. The bad news is that some military powers can use it to create a more advanced version of its existing pain beam.

Advantages of plasma antennas

1. An important advantage of plasma antenna over a conventional antenna is that the former is much lighter. Based on a set of patented beam-forming technologies, these high-performance electronically-steerable antennas are extremely lightweight and compact.

2. Free from mechanical parts, these maintenance-free plasma antennas are ideally suited for a wide range of wireless communications and sensing applications.

3. Plasma antennas have a number of potential advantages for antenna design. These are reconfigurable. When one plasma antenna is de-energised, the antenna reverts to a dielectric tube, and a second antenna can transmit through it. This allows using several large antennas stacked over each other instead of several small antennas placed next to each other. This results in better sensitivity and directivity.

4. When a plasma element is not energised it is difficult to detect it by radar. Even when it is energised, it is transparent to the transmissions above the plasma frequency, which falls in the microwave region.

5. Plasma elements can be energised and de-energised in seconds, which prevents signal degradation.

6. When a particular plasma element is not energised, its radiation does not affect nearby elements.

7. Plasma antenna can focus high-frequency radio waves that would

dissipate quickly if beamed by conventional arrays.

8. Plasma antennas boost wireless speeds. Such antennas could enable next-generation Wi-Fi that allows for super-fast wireless data transfers.

9. Solid-state plasma antennas deliver gigabit-bandwidth, and high-frequency plasma antenna could hold the key for economically viable super-fast wireless networking.

10. Plasma antennas might also be used to create low-cost radar arrays that could be mounted on cars to help them navigate in low-visibility conditions, or used to make directed, more focused and less bulky energy weapons.

11. Plasma antennas have developed an innovative range of selectable multi-beam antennas that meet the demands in today's wireless communication, defense and homeland security markets.

Limitations

1. The current hardware uses a wider range of frequencies so it's impractically massive to be used for mobile environments.

2. Plasma antennas are expensive and hard to manufacture.

3. High-frequency signals mean that antennas operating at higher frequencies couldn't penetrate walls like conventional Wi-Fi, so signals would have to be reflected throughout the buildings.

Plasma antennas could theoretically solve some of these problems because these can operate at a wider range of frequencies, but gas antennas are also more complex (and likely more expensive) than their silicon-diode counterparts, which are small enough to fit inside a cell phone.

With plasma antenna technology, there are kinks to iron out, but researchers and engineers are optimistic to make this promising technology commercially available in few years. ●

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